



Suman Kumar Raj

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EDUCATION

Degree/Certificate	Institute	CGPA/Percentage	Year
B.TECH + M.TECH	Indian Institute of Technology, Kharagpur	8.12	2020–2025
Class XII (CBSE)	S K P Vidya Vihar Mandroza Bhagalpur, Bihar	92%	2020
Class X (CBSE)	D P S Sabour Bhagalpur, Bihar	10	2018

PROJECTS

- **Multi-Agent Research Assistant**
Python, LangGraph, LangChain, OpenAI, Tavily, Wikipedia, LangSmith
 - o Built a multi-agent research system using **LangGraph** to decompose complex topics into specialized analyst perspectives with human-in-the-loop refinement.
 - o Implemented parallel information retrieval using **Tavily** and **Wikipedia**, enabling agents to gather and synthesize context from multiple sources.
 - o Designed a map-reduce workflow to parallelize analyst interviews and consolidate their outputs into a structured final research report.
- **MyTaskManager**
Python, LangGraph, LangChain, OpenAI, Pydantic, Trustcall
 - o Built a stateful AI task-management agent using LangGraph that maintains short-term conversational state and structured long-term memory across interactions.
 - o Designed separate **Profile**, **To-Do**, and **Instruction** memory with Pydantic schemas, enabling structured memory extraction, updates, and personalized task management.
 - o Implemented conditional agent routing and cyclic workflows to determine when memory updates are required, route operations to the appropriate memory node, and return control to the main agent.
- **Fingerprint Liveness Detection (Presentation Attack Detection)**
Python, TensorFlow, Keras, MobileNetV3-Small, OpenCV
 - o Built a **Presentation Attack Detection (PAD)** system using **MobileNetV3-Small** transfer learning to classify fingerprint images as **LIVE** or **SPOOF** for biometric authentication.
 - o Developed an end-to-end pipeline covering 70/15/15 dataset splitting, image augmentation, ImageNet normalization, model training, threshold calibration, and single-image inference.
 - o Calibrated the threshold against a target **BPCER** of **3%** and evaluated held-out test performance using **APCER**, **BPCER**, **ACER**, **EER**, **ROC-AUC**, **F1-score**, and **APCER–BPCER** trade-off analysis.
- **LendingClub Risk Prediction**
Python, Pandas, NumPy, Scikit-learn, TensorFlow/Keras
 - o Built an end-to-end credit risk prediction pipeline using TensorFlow/Keras on **396K+** LendingClub loan records, covering missing-value handling, categorical encoding, feature engineering, and MinMax scaling.
 - o Trained a deep neural network with ReLU activations, dropout, and sigmoid output to classify loans as fully paid or defaulted.
 - o Achieved a **0.93 F1-score** on the test set, effectively handling significant class imbalance in loan default prediction.

SKILLS SUMMARY

- **Generative AI:** LangChain, LangGraph, Agentic AI, Memory Systems, Multi-Agent Systems, LLM Orchestration, RAG, VectorDB, GraphDB, Prompt Engineering, Tool & Function Calling, Structured Outputs, LLM Evaluation.
- **Machine Learning:** Classification, Regression, EDA, NLP, Neural Networks.
- **Programming Languages:** Python, SQL
- **Frameworks and Libraries:** TensorFlow, Keras, Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, TensorBoard.
- **Platforms & Tools:** Git, GitHub, LangSmith, VS Code, Jupyter Notebook, Google Colab, Kaggle.
- **Data Structures and Algorithms, DBMS, Object-Oriented Programming, MLOps, Probability and Statistics**

EXPERIENCE

- **Mentor | AlgoZenith** [Oct'22 - May'24]
 - o Mentored **500+** students in Data Structures and Algorithms, helping improve problem-solving and interview preparation.
 - o Guided students through coding challenges, optimization techniques, and interview-focused problem-solving approaches.
- **Teaching Assistant | Indian Institute of Technology, Kharagpur** [Aug'23 - May'25]
 - o Assisted in conducting tutorials and evaluating assignments for undergraduate courses at IIT Kharagpur.
 - o Facilitated doubt-clearing sessions, helping students build stronger understanding of core concepts and analytical techniques.

AWARDS AND ACHIEVEMENTS

- Qualified **GATE 2024** with All India Rank (AIR) **64**.
- Received the prestigious **Prof. J.P. Ghose Memorial Award (2022–23)** for exceptional academic performance.
- **Gold Medal** recipient and distinguished participant in the Gymkhana Championship Choreography (2021–2022).

OPEN SOURCE

- **CodeBeat — Open-Source Python Package | PyPI |**
 - o Created a Python package for line-by-line function tracing and performance profiling, measuring execution-time statistics across repeated runs to identify code bottlenecks.